



Forsta Custom Dialer API. v.1.0

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Introduction

The Forsta Custom Dialer API allows to attach a custom dialer directly to the CATI Interviewer Console.

Quick Start

To attach a custom dialer, you need to get your dialer implementation in the Dynamic Link Library (dll) format. Instructions for creating implementation in such format can be found below in the *Create your own dialer* section. For test purposes, you can use supplied demo project and it's **CustomDialerDemo.dll**.

After you have received the .dll file with the dialer, complete the following steps:

1. Put the .dll file in any convenient folder and copy its full path.
For example, if you choose the supplied demo project, take CustomDialerDemo.dll and put it to `C:\Users\MyUser\Desktop\CustomDLL` folder, the full path will look like this `C:\Users\MyUser\Desktop\CustomDLL\CustomDialerDemo.dll`.
2. Open the %PROGRAMDATA%\Confirmit\CATI folder and find the file Settings.config. Open it in a text editor and make changes according to the *Configuration* section of this manual.
3. Restart CATI Console

Create your own dialer

To create your own dialer in the .dll format you need:

1. Create a .NET Framework 4.6.2 class library (dll) project.
2. Add a reference to **Confirmit.CATI.Console.LightweightTelephony.dll** in this project.
3. Create a class implementing **ICustomDialer** interface.
4. Build the project and take the dll version from the **bin** folder

You can see an example implementation in the supplied demo project.

Configuration

When you open the *settings.config* file in a text editor, you will see the current settings for the CATI Interviewer Console.

You need to configure the *CustomDialerPath* and *CustomDialerType* settings as follows:

```
<Settings>
...
<CustomDialerPath>FullPath</CustomDialerPath>
<CustomDialerType>CustomDialer_NameSpace.CustomDialer_ClassName</CustomDialerType>
</Settings>
```

where *FullPath* is the full path to the .dll file you saved in step 1 of the *Quick Start* section,

and *CustomDialer_NameSpace.CustomDialer_ClassName* is the full name of the class and namespace that you used in your dialer.

For example, for the supplied demo project and example path from step 1 of the *Quick Start* section, settings will look like this:

```
<Settings>
...
<CustomDialerPath>C:\Users\MyUser\Desktop\CustomDLL\CustomDialerDemo.dll</CustomDialerPath>
```

```
<CustomDialerType> CustomDialerDemo.CustomDialerDemo</CustomDialerType>
</Settings>
```

Demo project

The supplied **CustomDialerDemo** project can be used as a reference. It does not do real dialing but simulates connected and not connected calls on random basis.

ICustomDialer Interface

```
public interface ICustomDialer
{
    void Dial(string phoneNumber);
    void HangUp();
    event EventHandler<CallStatusChangedEventArgs> CallStatusChanged;
}
```

The interface is pretty simple. CATI Interviewer Console calls the **Dial()** method passing phone number as a parameter. Once dialing is completed (connected or not) the custom dll code must trigger the **CallStatusChanged** event as follows:

```
CallStatusChanged(this, new CallStatusChangedEventArgs {CustomCallStatus = callOutcome});
```

Note that it is very important to trigger the event because otherwise the CATI Console will be suspended at the moment the dial screen is displayed.

If a custom dialer does not support the return of a call outcome status you may just trigger an event with the status “Non automatically sensed” after receiving the Dial command and then handle it appropriately in a survey script.

The **Hangup()** method is called by the Console once a call is finished. Custom code should disconnect the call and clear all the resources allocated to handle the call.

Data Types

CallStatusChangedEventArgs

```
public class CallStatusChangedEventArgs : EventArgs
{
    public CustomCallOutcome CustomCallStatus;
}
```

CustomCallOutcome

```
public enum CustomCallOutcome
{
    Connected=0,
    Busy=2,
    NoReply=3,
    Refusal=5,
    Terminated=6,
    AnswerPhone=7,
    Modem=8,
    Fax=9,
    Congestion=10,
    Unobtainable=11,
    Nuisance=12,
    Completed=13,
    Screened=14,
    ReturnedNotDialled=15,
```

```
NotAutomaticallyDialled=18,  
ReturnedDiallerExpired = 25,  
InterruptedBySystem=26,  
Stopped=28,  
TelephonyFailure=29,  
Error=30  
}
```